# Work along a displacement

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Calculate work and identify when a force does no work.

**SUCCESS CRITERIA - I can...**

1. Identify the force component along displacement.
2. Use meters and newtons to obtain joules.
3. Distinguish effort from mechanical work on an object.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

Keep impulse fixed, multiply target mass by 3 and lead time by 5. Compare the modeled transverse shift using an equation.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** W = Fd.
Constant force parallel to displacement; W is work. No displacement means no work on the object by that force.

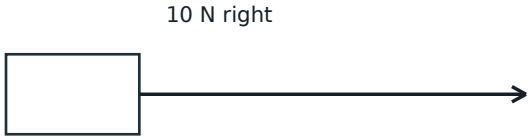
**Worked reasoning:** A constant 5 N pull acts along a 3 m displacement. Work is Fd=15 J. The object must move for this force to do mechanical work.

**Your turn:** Draw the force and displacement arrows. Show the 15 J as area on a force-versus-position graph.

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

A box moves 4 m right while a 10 N force acts right. Draw force and displacement arrows; predict the sign of work.




4 m right

The rectangle is the box. Add the requested force and displacement labels.

**Marzano check - circle:** ① ② ③

## 04 Solve and justify

A constant 18 N force acts along a 5 m displacement. Friction is 6 N opposite. Find work by each force and net work.

**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

# Work along a displacement

**antocciphysics**



Continue down each column.

### 05 Analyze the reasoning

A student holds a heavy bag still. Is the upward support force doing mechanical work on the bag during the hold? Explain without denying the student feels tired.

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A 4 kg object slows from 8 to 4 m/s in 1.5 s. Find impulse and average net force. Repeat the force calculation for the same velocity change over 3 s.

**Marzano check - circle:** 1 2 3

### 07 Exit check

A force does 72 J over 3 m parallel to motion. Find force. If displacement were zero, what work would that force do?

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A varying force rises linearly from 2 to 8 N over 4 m. Find work from a graph and compare with using only the final force.

**Marzano check - circle:** 1 2 3